



HTTP Web Server User Manual

This manual provides a practical overview, setup guide, and execution walkthrough for running the standard C++ Winsock HTTP Server application.

Technical Specifications

Feature	Specification
Language & Standard	C++11 or higher
Networking Library	Winsock2 (ws2_32.lib)
Supported OS	Windows (7, 8, 10, 11)
Protocol	HTTP/1.1 (TCP/IP)
Concurrency	Single-threaded synchronous
Default File	Index.html

System Requirements & Dependencies

- **Compiler:** MinGW (GCC/G++), Microsoft Visual C++ (MSVC), or Clang.
- **Libraries:** Winsock2 library (ws2_32.lib).
- **Header Files:**
 - `<winsock2.h>`
 - `<iostream>`, `<fstream>`, `<string>`, `<conio.h>`
 - Local source file: `titleScreen.cpp` (must contain a callable `title()` function).

Compilation Instructions

Using GCC / MinGW (Command Line)

Run the following command in your terminal to compile and link the ws2_32 Winsock library:

Bash

```
g++ main.cpp -o server.exe -lws2_32
```

Using Visual Studio (MSVC)

1. Open the project in Visual Studio.
2. Ensure `#pragma comment(lib, "ws2_32.lib")` remains at the top of the file.
3. Build using `Ctrl + Shift + B` or select **Build > Build Solution**.

Directory Structure Prerequisites

Before launching the server, organize your target directory to contain an `Index.html` file.

Plaintext

Project_Directory/

|

|— server.exe

|— titleScreen.cpp

|— site/

|— Index.html

Running the Server

1. **Launch the Executable:** Run `server.exe` from your terminal or double-click the application.
2. **Specify Site Path:** Provide the relative or absolute directory path containing your HTML document.
 - Example input: `./site`
3. **Specify Port Number:** Input a port above 1024 to avoid administrator privilege conflicts (e.g., 8080).

Console Example Session

example : `path/to/your/site`

Try : `./site`

Enter path to host : `./site`

Enter port : 8080

server running...

Access in Browser: Open any Web Browser and navigate to:

- `http://localhost:8080` or `[http://127.0.0.1:8080](http://127.0.0.1:8080)`

Code Architecture & Runtime Flow

[Start Program] -> [Print Title] -> [Prompt Path & Port]

|



[Initialize Winsock (WSAStartup)]

|



[Create Socket] -> [Bind to IP/Port] -> [Listen for Connections]

|



[Main Loop (while 1)]

- └─► [Accept Connection]
- └─► [Read Client HTTP Request]
- └─► [Read Index.html]
- └─► [Send HTTP 200 OK Response]
- └─► [Close Client Socket]

Operational Behavior & Edge Cases

- **Static HTML Delivery:** The server attempts to read Index.html dynamically during each incoming connection request.
- **Single-Serve Limitation:** File reading is initialized outside the main loop (file(...)). The file state remains loaded in memory and does not reload dynamically between requests unless the file pointer is reset.
- **Capitalization Sensitivity:** Ensure the target HTML document matches the exact casing expected by the program (Index.html).
- **Connection Lifecycle:** The server utilizes short-lived TCP connections (Connection: close). The client socket closes immediately after sending the response payload.

Common Error Codes & Troubleshooting

Error Code / Message	Cause	Solution
WSA Error 10048	Port is already in use by another application.	Use a different port (e.g., 8081 or 9000).
WSA Error 10061	Server is not listening or firewall blocked socket.	Allow server.exe through Windows Defender Firewall.
Empty Page Delivered	Index.html missing or incorrect directory path.	Verify file casing and check that Index.html exists in the target folder.
Compilation error: undefined reference to WSASStartup	ws2_32.lib was not linked correctly.	Include -lws2_32 when compiling with g++.